

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Faculty of Engineering and Technology

First Semester of M.Tech. (CE/IT)

Dec-Jan -2013

CE704 Soft Computing

Date: 04.01.2014, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answer the following questions.

- (a) What are the objectives of soft computing? How it is differ from hard computing? Briefly mention the application area of soft computing? [03]
- (b) Differentiate between feed-forward and feed-back network. [02]
- (c) Explain terms: Crisp Logic, Fuzzy Logic. [02]

- Q - 2
- (a) "Neural Network always learns faster than other Classifier." Justify. [04]
 - (b) What is Learning? How ANN do learning? Differentiate between Competitive Learning and Supervised Learning. [07]

OR

- (b) How learning rate play important role in learning? How can the training of neural network be improved? Differentiate inverse learning and simple learning. [07]
- (c) Explain Feature map? Discuss Kohonen Self Organization Map(KSOM) in brief? [03]

OR

- (c) Which are the factors influencing the Back Propagation training? Mention the heuristics which will significantly improve the performance of Back propagation algorithm. [03]

- Q - 3
- (a) Explain terms with respect to fuzzy logic: Membership function, Support, Core, Cross over point, Sington. [05]
 - (b) How fuzzification is differing from defuzzification? How to do fuzzification? List and explain different defuzzification methods with example. [06]

OR

- (b) The training data for a particular problem is given in below table. Use BPN to train the data. Show step by step output to input, hidden and output neurons as well as error. How the weights W and V are modified? Use learning Coefficient and momentum factor as 0.6 and 0.9 respectively. [06]

Input				Output
0.16	0.12	0.08	0.04	0.2
0.12	0.24	0.16	0.08	0.4
0.08	0.16	0.24	0.12	0.6
0.04	0.08	0.12	0.16	0.8

- (c) What is Generalized Modus Ponon and Generalized Modus Tonon? Explain Modus Ponon, Modus Tonon, if-then rule? [03]

OR

- (c) List various Fuzzy membership operations with their equation. Let O and P be two fuzzy sets of old and poor people. What is the membership function value of a person being old and poor if the member grade of a person being old is 0.8 and being poor is 0.7? [03]

SECTION – II

Q-4 Answer the following questions.

- (a) While learning, explain how generational Cycle works with Genetic algorithm? Discuss briefly. [02]
- (b) How can genetic algorithm solve the weight determination problem of neural network? [02]
- (c) Why Population is required in GA? Which is the operator applied first to the population? [03]
- Q-5 (a) How can Fitness functions find out for any optimization problem? Explain Fitness Function in Genetic algorithm in details. [06]
- (b) How can genetic algorithm be controlled by Fuzzy Logic? [04]

OR

- (b) Can we solve Traveling Sales Man Problem using Genetic Algorithm? How? Write the steps in brief. [04]
- (c) How specialized learning can improve the learning process of Hybrid approach? [04]

OR

- (c) How can genetic algorithm solve the weight determination problem of neural Network [04]
- Q - 6 (a) List the types of hybrid system and its application domain, where hybrid system are used. [04]
- (b) Write the hybridization steps for optimization problem using “Genetic-Fuzzy-Neural Network”. [05]

OR

- (b) How Neuro-Fuzzy modeling approach can be applied to any optimization problem? List and explain briefly at least two application domain of Neuro-Fuzzy Hybrid system. [05]
- (c) In a three variable problem the following variable bounds are specified [05]
 $-6 < x < 12$; $0.002 \leq y \leq 0.004$; $10^4 \leq z \leq 10^5$
What should be the minimum string length of any point (x,y,z) coded in ternary string (with three alleles 0,1,2) are used to achieve the following accuracy in the solution:
1. Two significant digit
2. Three significant digits.

OR

- (c) Select any application domain of Fuzzy system, design fuzzy rule based system to infer output fuzzy value. Take suitable fuzzy set of domain as input and output. Defuzzification of input, apply designed rule base and then do defuzzification of fuzzy output for particular value of input. [05]